FIITJEE

ALL INDIA TEST SERIES

FULL TEST – III

JEE (Advanced)-2019

PAPER –2

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-D.**

1. ***Section-A (01 – 06, 19 – 24, 37 – 42)*** *contains 18 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen, both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–2** In all other cases.*

   Section-A (07 – 10, 25 – 28, 43 – 46) *contains 12 questions. Each question has* ***TWO (02)*** *matching lists:* ***LIST-I*** *and* ***LIST-II****.* ***FOUR*** *options are given representing matching of elements from* ***LIST-I*** *and* ***LIST-II****.* ***ONLY ONE*** *of these four options corresponds to a correct matching.*
   For each question, choose the option corresponding to the correct matching.
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-D (11 – 18, 29 – 36, 47 – 54)*** *contains 24 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

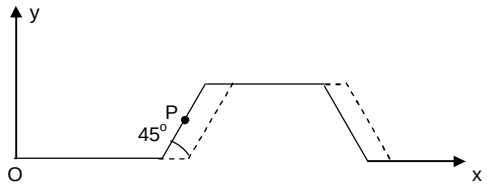
PART – I

SECTION – A

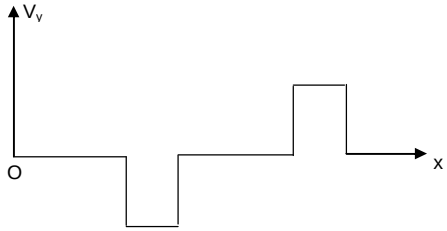
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. The figure shows a taut string, initially aligned with the x-axis, which carries a waveform travelling towards the right without any change its shape. The bold line and the dotted line indicate the displacement of the string (i.e., shape of the waveform) at time = t second and at time = (t + dt) second, respectively. The speed of particle at point P at t = 2 sec is 1 cm/s. Choose the correct statement(s) from the following (assume that dt is positive).

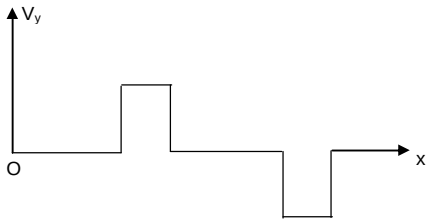


(A)



Particle velocity as a function of position

(B)



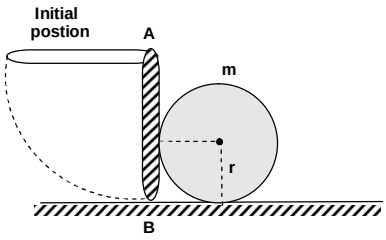
Particle velocity as a function of position

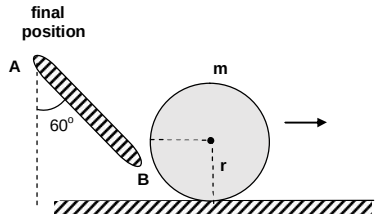
(C) Speed of waveform is 1 cm/s and it is moving along positive x-axis

(D) Speed of waveform is 1 cm/s and it is moving along negative x-axis

Space for Rough work

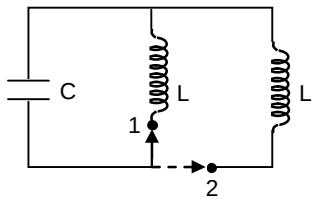
2. A uniform rod AB of mass M is attached to a hinge at one end A, and released from rest from the horizontal position. The rod rotates about A, and when it reaches the vertical position the rod strikes a sphere of mass m and radius r initially at rest on the smooth horizontal surface as shown in the adjacent figure. The impact is along the horizontal direction and perfectly elastic. If at the moment of impact, the lowest end of the rod is very close to the smooth horizontal surface. After the impact, the sphere moves along the horizontal and the rod, subsequently rises to a maximum of 60° with the vertical. Choose the correct statement(s) form the following, taking into account the information given above. The length of the rod equals $\sqrt{2r}$. ($r = \frac{6\sqrt{2}}{10}$ m)





(A) The ratio M/m is 3/2
(B) The ratio m/M is 2/3
(C) The speed of the sphere just after collision is 6 m/s
(D) The speed of the sphere just after collision is 3 m/s

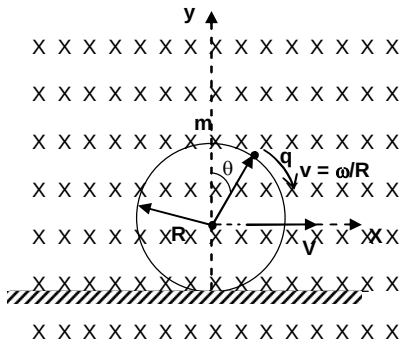
3. A capacitor with charge Q on it is connected to an inductor L as shown in diagram at t = 0. When the switch is flipped from position 1 to 2, the current in the circuit is observed to be at half of its maximum value. Then



(A) Charge on capacitor at the time was Q/2
(B) Charge on capacitor at that time was $\frac{Q\sqrt{3}}{2}$
(C) Total energy lost is $\frac{Q^2}{4C}$
(D) Total energy lost is $\frac{Q^2}{8C}$

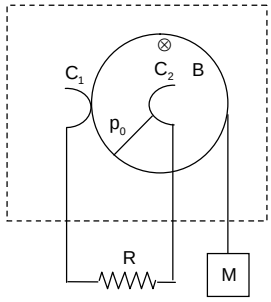
Space for Rough work

4. A ring of mass m and radius R is set into pure rolling on horizontal rough surface, in a uniform magnetic field of strength B – as shown in the figure. A point charge q of negligible mass is attached to rolling ring. Friction is sufficient so that it does not slip at any point of its motion. (θ is measured in clockwise from positive y-axis)



(A) Ring will continue to move with constant velocity.
(B) The value of friction acting on ring is Bqv cos θ.
(C) The value of friction acting on ring is Bqv sin θ.
(D) Ring will lose contact with ground if v is greater than $\left(\frac{mg}{2qB}\right)$.

5. Consider a perfectly conducting disc of radius r_o in a constant magnetic field B perpendicular to the plane of the disk sliding contacts are provided at the edge of the disc (C_1) and its axle (C_2). After long time angular velocity will become constant and becomes equal to ω & current in resistance is i, then



(A) $i = \frac{B\omega r_o^2}{R}$ (B) $i = \frac{B\omega r_o^2}{2R}$

(C) $\omega = \frac{4MgR}{B^2 r_o^3}$ (D) $\omega = \frac{2MgR}{B^2 r_o^3}$

6. Some successive frequencies of an ideal organ pipe are measured as 150 Hz, 250 Hz, 350 Hz. Velocity of sound is 340 m/s.
(A) it must be an open organ pipe of length 3.4 m
(B) it must be an open organ pipe of length 6.8 m
(C) it must be an closed organ pipe of length 1.7 m
(D) it must be an closed organ pipe of length 3.4 m

Space for Rough work

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST - I** and **LIST - II**. **FOUR** options are given representing matching of elements from **LIST - I** and **LIST - II**. **ONLY ONE** of these four options corresponds to a correct matching.

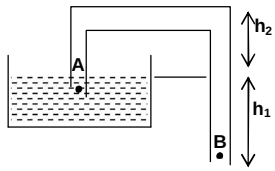
7. In closed organ pipe,

List –I		**List –II**	
(P)	Third overtone frequency is times fundamental frequency then n is	(1)	3
(Q)	Number of nodes in second overtone	(2)	4
(R)	Number of antinodes in second overtone	(3)	5
(S)	n^{th} harmonic does not exist whose n is	(4)	7

Codes:

	P	Q	R	S
(A)	4	1	2	3
(B)	4	1	1	2
(C)	3	1	2	2
(D)	3	4	1	2

8. A siphon transfers liquid from high level to low lend as shown. If P is density of fig. P_o is atmospheric pressure. V_a in redo. At pt. A and V_B at B then.



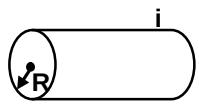
List –I		**List –II**	
(P)	$(h_1 + h_2) - \frac{P_o}{P_g}$	(1)	= 1
(Q)	$V_A - V_B$	(2)	< 0
(R)	h_1	(3)	> 0
(S)	P_A-P_B	(4)	= 0

Codes:

	P	Q	R	S
(A)	2	1	4	1
(B)	2	4	1	2
(C)	2	1	3	4
(D)	1	2	3	4

Space for Rough work

9. A long straight conductor of radius R carries a current I as shown. The current density j varies as a function of radius according to j = br where b is a constant. Match the quantities in List – I with those in List – II.



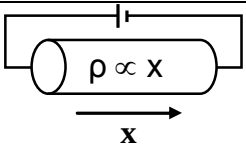
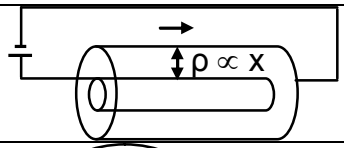
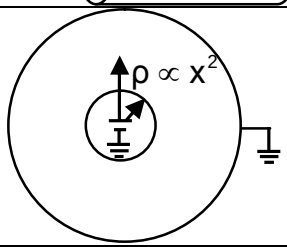
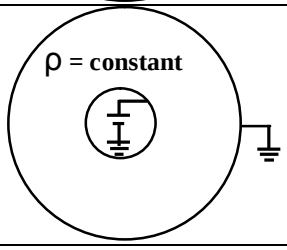
List –I		**List –II**	
(P)	B_{inside}	(1)	Maximum
(Q)	$B_{surface}$	(2)	$\frac{\mu_0 br^2}{3}$
(R)	$B_{outside}$	(3)	$\frac{2\pi br^3}{3}$
(S)	I_{inside}	(4)	$\frac{\mu_0 bR^3}{3r}$
		(5)	Continuous

Codes:

	P	Q	R	S
(A)	1	2	4,5	3
(B)	2	1,5	4	3
(C)	3	5	2	3,5
(D)	4,5	1	4	3

Space for Rough work

10. List – I has some conductor across which battery is connected as shown. Variation of resistivity ρ is also indicated. Which of the quantities in List – II remains constant throughout the volume of conductor.

List –I		List –II	
(P)		(1)	Magnitude of electric field
(Q)		(2)	Magnitude of current density
(R)		(3)	Power dissipated per unit volume
(S)		(4)	Drift speed of free electron
		(5)	Electric current

Codes:

	P	Q	R	S
(A)	2,4	4	5	2,5
(B)	4	1,3	1	4,5
(C)	1,4,5	2,4	2,4,5	3,5
(D)	2,4,5	2,4,5	1,5	5

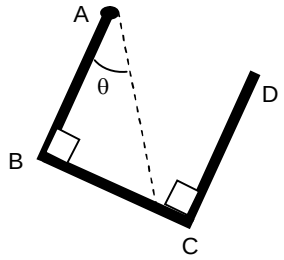
Space for Rough work

SECTION – D

(Numerical Answer Type)

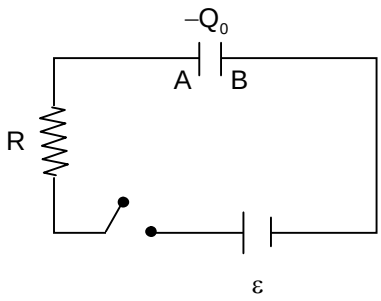
This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

11. Three identical rods are hinged at point A as shown.
The angle made by rod AB with vertical is θ. Find the value of $5 \times \sin\theta$

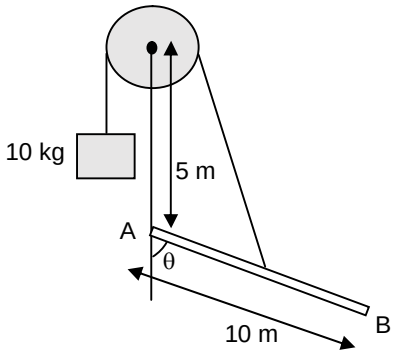


12. The figure shows a RC circuit with a parallel plate capacitor. Before switching on the circuit, plate A of the capacitor was given a charge $-Q_0$ while plate B had no net charge. Now, at t = 0, the circuit is switched on. Time (in seconds) will elapse before the net charge on plate A becomes zero is K. Value of 125K is

Take $C = 1\,\mu F, Q_0 = 1\,mC, \varepsilon = 1000$ V and $R = \dfrac{2\times10^6}{\ln(3)}\Omega$

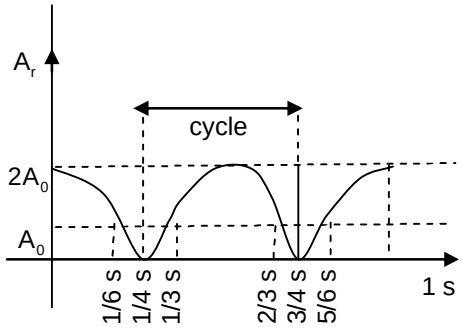


13. A uniform rod AB of mass 5 kg and length 10m is hinged at A. A block of mass 10 kg is hanging with a string and other end of the string is attached to a middle point of the rod passes over a smooth pulley as shown in the figure. Find the value of $365\left(\cos\dfrac{\theta}{2}\right)$.

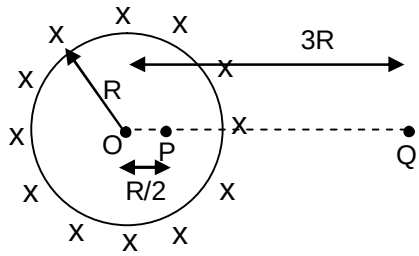


Space for Rough work

14. Two sound waves of frequencies 100 Hz and 102 Hz and having same amplitude 'A' are interfering. A stationary detector, which can detect waves of amplitude greater than or equal to A, in a given time interval of 12 seconds, find the total duration in which detector is active.



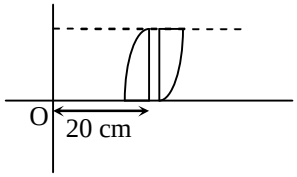
15. Conducting shell of radius R carrying charge Q, another charge Q is placed at 3R distance from centre of shell. If the potential at point P due to charge on shell is $\frac{11}{3n}\frac{KQ}{R}$, Then the value of 20n is



16. A liquid is kept in a cylindrical vessel. When the vessel is rotated about its axis, the liquid rises at its sides. If the radius of the vessel is 0.05 m and the speed of rotation is 2 revolutions per second, the difference in the heights (in cm) of the liquid at the centre and at the sides of the vessel is (take $g = 10\ ms^{-2}$ and $\pi^2 = 10$)

17. If two spherical bubbles of same radius 2 m. and same liquid coalesece to form a common plane surface, then find the gap between the centres of the bubbles is n. Then the value of 393n is

18. A biconvex thin lens of focal length 30 cm and aperture diameter 10 cm is broken into four identical pieces. Two of the pieces are placed in a coordinate system as shown in the diagram. Find the magnitude of the y coordinate of the image if an object is placed at origin.



Space for Rough work

Chemistry

PART – II

SECTION – A

(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. The solubility of $CH_3COOAg\left(K_{sp} = X\right)$ in a buffer solution of $pH = Y\left(Y < 7\right)$

(A) increases as the value of Y decreases
(B) decreases as the value of Y decreases
(C) will be greater if the K_a value of the CH_3COOH is greater
(D) will be smaller if the K_a value of the CH_3COOH is greater

20. For the gas phase reaction,

$C_2H_4 + H_2 \rightleftharpoons C_2H_6;\ \Delta H = -137$ kJ mol^{-1}

Carried out in a closed vessel, the equilibrium concentration of C_2H_4 can be increased by

(A) Increasing the temperature
(B) Decreasing the pressure
(C) Removing some C_2H_6
(D) Adding some H_2

21. A vessel at 1000 K contains CO_2 with a pressure of 0.5 atm. Some of the CO_2 is converted into CO on adding graphite. The total pressure at equilibrium was found to be 0.8 atm.

(A) The equilibrium is achieved as soon as $Q = K$, K value = 0.18 atm^2
(B) The equilibrium is achieved as soon as Q = K, K value = 0.18 atm
(C) The $p_{CO_2} = 0.2$ atm at equilibrium
(D) The $p_{CO} = 0.6$ atm at equilibrium

22. Potassium chromate solution is added to aqueous solutions of metal nitrate. The yellow precipitate thus obtained are insoluble in acetic acid. These are subjected to flame test, flame colour of individual ppt. is/are

(A) Lilac
(B) Apple green
(C) Crimson red
(D) Blue

Space for Rough work

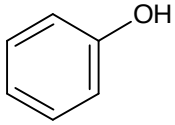
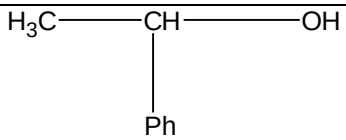
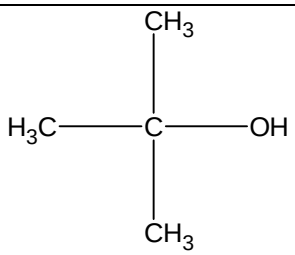
23. Synergic bonding in metal carbonyl complexes [M = Metal]:
(A) decreases C – O bond strength
(B) increases M – C bond length
(C) increases bond order for M – C bond
(D) increases paramagnetic of complex

24. $\left[M(NH_3)_2(py)_2(NO_2)_2\right] \xrightarrow[-NO_2]{+NH_3} \left[M(NH_3)_3(py)_2(NO_2)\right]$ Difference between the possible geometrical isomers of reactant and product is:
(A) 3 (B) 4
(C) 2 (D) 0

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST - I** and **LIST - II**. **FOUR** options are given representing matching of elements from **LIST - I** and **LIST - II**. **ONLY ONE** of these four options corresponds to a correct matching.

25. Match List I with List II and choose the correct answer using the codes given below:

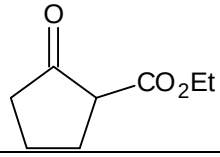
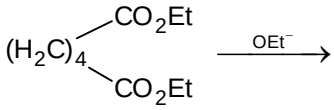
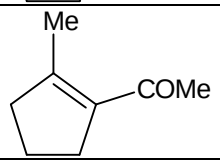
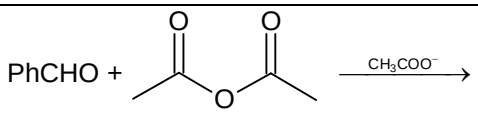
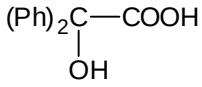
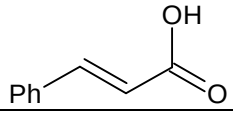
List –I		**List –II**	
(P)		(1)	White turbidity with $HCl / ZnCl_2$
(Q)	CH_3CH_2OH	(2)	Violet colour with $FeCl_3$
(R)		(3)	Colour change with $Na_2Cr_2O_7, H^+$
(S)		(4)	I_2 / OH^-, gives bright yellow ppt

Codes:

	P	Q	R	S
(A)	1,2	3,4	1,4	2
(B)	1,2,3	1	3,4	1,2,3
(C)	2	3,4	1,3,4	1
(D)	2	1,3,4	3,4	2

Space for Rough work

26. Match List I (reaction) with List II (product) and then select the correct answer from the codes given below the lists:

List –I (Reactions)		List –II (Product)	
(P)	$MeCO(CH_2)_4COMe \xrightarrow[(ii)\ \Delta]{(i)\ OH^-}$	(1)	
(Q)		(2)	
(R)		(3)	
(S)	$PhCOPh \xrightarrow[H_3O^+]{HCN}$	(4)	

Codes:

	P	Q	R	S
(A)	1	2	3	4
(B)	1	2	4	3
(C)	2	1	3	4
(D)	2	1	4	3

27. Match List I with List II and then select the correct answer from the codes given below the lists:

List –I (Reactants)		List –II (Product)	
(P)	Benzaldehyde + $NaBH_4$	(1)	Benzoic acid
(Q)	Benzoyl chloride + H_2 + Pd + $BaSO_4$ + Δ	(2)	Benzyl amine
(R)	Benzyl alcohol + H_2CrO_4	(3)	Benzyl alcohol
(S)	Benzamide + $LiAlH_4$	(4)	Benzaldehyde

Codes:

	P	Q	R	S
(A)	3	4	1	2
(B)	2	3	1	4
(C)	3	4	2	1
(D)	3	1	4	2

Space for Rough work

28. Match List –I (Group when attached to phenyl ring) with List –II (Effect shown) and select the correct answer using the code given below lists:

List –I		List –II	
(P)	$-Cl$	(1)	+m and +I
(Q)	$-O^-$	(2)	+m and –I
(R)	$-CH_2-CH_3$	(3)	Only –I
(S)	$-\overset{\oplus}{N}H_3$	(4)	+I and Hyperconjugation

Codes:

	P	Q	R	S
(A)	1	3	2	4
(B)	3	4	2	1
(C)	4	2	3	1
(D)	2	1	4	3

SECTION – D

(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

29. The angular momentum of an electron in a Bohr's orbit of H atom is 4.2178×10^{-34} Kg m^2 / sec . The wavelength of spectral line emitted when electron falls from this level to next lower level is $x\times10^{-4}$ cm. Find the value of x.

30. $\left[(C_2H_5)_x Al\right]$ and $TiCl_4$ form the basis of Zeigler-Natta type heterogeneous catalyst. What is x?

31. In the structure of XeF_2. $2SbF_5$ in solid state how many types of Xe-F bond lengths are present?

32. How many of the following are pseudohalogens?
Azidocarbon disulphide, Isocyanide ions, Fluminate ion, Tellurocynante ion, Oxyeyanogen. Selerocyanogen.

33. When $K_2Cr_2O_7$ reacts with HCl, in how many products chlorine exits as chloride ion?

34. What is the oxidation number of Cl_2 in final product formed due to oxidation in it when Cl_2 reacts with hot NaOH?

35. How many of the following oxides are neutral oxides?
$Cl_2O_7, SO_3, SrO, H_2O, CO, N_2O_5, N_2O, P_2O_5$ and No

36. Number of lone pair-bond pair repulsion at 90° are (P) in I_3^-

Space for Rough work

Mathematics

PART – III

SECTION – A

(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. Let $f : [0, 1] \to [0, \infty)$ be a differentiable with decreasing first derivative & $f(0) = 0$, then $\int_0^1 \frac{dx}{f^2(x)+1}$ is

(A) $< \frac{\tan^{-1} f(1)}{f'(1)}$ (B) $< \frac{f(1)}{f'(1)}$

(C) $> \frac{\tan^{-1} f'(1)}{f'(1)}$ (D) $\le \frac{f(1)}{f'(1)}$

38. Let $f : R \to (0, \infty)$ be such that $f(x) + \frac{e^{x+x^2}}{f(x)} \le e^x + e^{x^2}$ $\forall$ $x > 0$, then $\lim_{x\to 1} f(x)$ is equal to

(A) y where $y = \sqrt{\lim_{n\to\infty} n^{-n^2}} \left[(n+1)\left(n+\frac{1}{2}\right)\left(n+\frac{1}{2^2}\right).....\left(n+\frac{1}{2^{n-1}}\right)\right]$

(B) p^3 where $p^2 = \lim_{x\to 0}\left(\frac{1+5x^2}{1+3x^2}\right)^{\frac{1}{x^2}}$

(C) t where $\frac{1}{t} = \lim_{x\to 0} \frac{\tan^{-1} x - \sin^{-1} x}{\sin^3 x}$

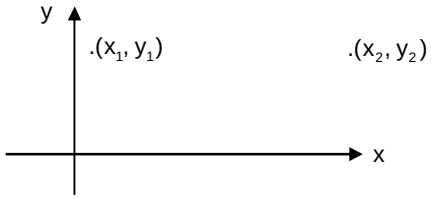
(D) r where $r = \frac{3}{5_e} \lim_{n\to\infty} \sum_{K=1}^{n} \frac{K^5 + 6K^2 + 11K + 5}{(K+3)!}$

Space for Rough work

39. If $P(x, y)$ means to make the point (x, y) on $x - y$ plane dark.



$P(x_1, y_1) + P(x_2, y_2)$ means to make both the points (x_1, y_1) & (x_2, y_2) as dark on $x - y$ plane.



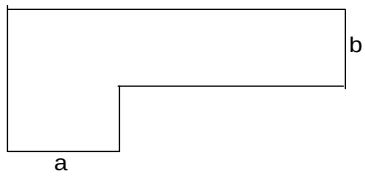
(A) $\lim\limits_{n\to\infty} \sum\limits_{r=1}^{2n} P\left(1, \frac{r}{n}\right)$ represents a line segment

(B) $\lim\limits_{n\to\infty} \sum\limits_{r=1}^{2n} P\left(\frac{r}{n}, \left(\frac{r}{n}\right)^2\right)$ represents a parabolic segment

(C) $\lim\limits_{n\to\infty} \sum\limits_{r=1}^{2n} P\left(1, \frac{r}{n}\right)$ represents a circle

(D) $\lim\limits_{n\to\infty} \sum\limits_{r=1}^{m} P\left(\frac{r}{n}, \left(\frac{r}{n}\right)^2\right)$ represents an ellipse

40. A corridor housing the shape as shown below is 'a' unit wide in one of its wings & b units wide in the other stick is moved from one of the wings to the other (It is assumed that the thickness of the stick is negligible & during the motion the stick stays horizontal)



(A) The length of longest stick that can move is $(a^{2/3} + b^{2/3})^{3/2}$

(B) The length of longest stick that can move is $(a^{3/2} + b^{3/2})^{2/3}$

(C) The longest stick makes L α with one of its walls & is tangent to vertex 0 of the inside might angle of the corridor then $\tan\alpha = (b/a)^{1/3}$.

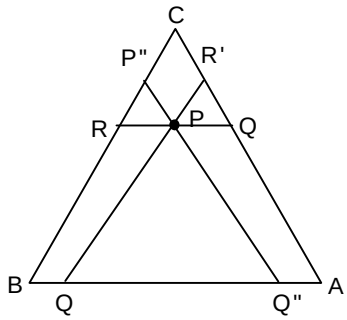
(D) None of these

Space for Rough work

41. Let A, B, C, D be non-singular real matrices such that $A^T = BCD, B^T = CDA, C^T = DAB$ & $D^T = ABC$ for the matrix $M = ABCD$, then find M^{2018}.

(A) M (B) M^2
(C) M^3 (D) M^4

42. In a Δ ABC, AB = 85, BC = 90 & AC = 102. P is an interior pt & line segments are drawn through P parallel to the sides of the triangle. If these three line segments (as shown in the diagram by RQ, R'Q', R"Q") are of equal length d, then $\frac{5d}{34}$ is equal to



(A) 9 (B) 6
(C) 3 (D) 12

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST - I** and **LIST - II**. **FOUR** options are given representing matching of elements from **LIST - I** and **LIST - II**. **ONLY ONE** of these four options corresponds to a correct matching.

43. Let f(x) and g(x) be two distinct functions such that $f^{-1}\left(\frac{f(x)+f(y)}{2}\right) = g^{-1}\left(\frac{g(x)+g(y)}{2}\right)$, then:

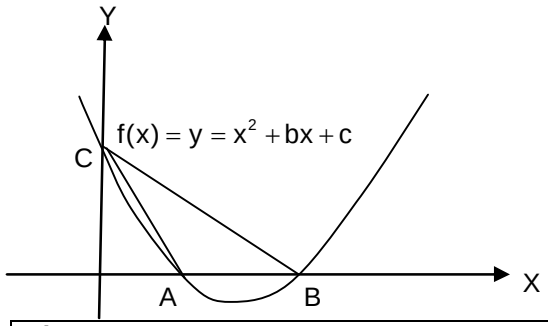
List –I		List –II	
(P)	If $f(x) = x^{10}\ \forall\ x \geq 0$ & $g(1) = 2, g(0) = 1$, then	(1)	No. of solutions of $g(x) = 0$ is 0
(Q)	If $f(x) = x^5$ & $g(1) = 7$ & $g(0) = 2$	(2)	No. of solutions of $g(x) = 0$ is infinite
(R)	If $f(x) = \tan x$ & $g(0) = 1$ & $g(\pi/4) = \sqrt{3}$	(3)	No. of solutions of $g(x) = 0$ is 1
(S)	If $f(x) = x^3$ & $g(0) = 5$ & $g(1) = 2$	(4)	No. of solutions of $g(x) = 0$ is 2

Codes:

	P	Q	R	S
(A)	2	3	2	1
(B)	1	3	2	3
(C)	4	1	3	2
(D)	1	2	4	3

Space for Rough work

44. In the given figure ΔOBC is an isosceles right triangle in which AC is median:



List –I		List –II	
(P)	The equation whose roots are $(\alpha+\beta)$ & $(\alpha-\beta)$, where α, β $(\alpha > \beta)$ are roots of $x^2+bx+c=0$ is $x^2-px+3=0$, then p is	(1)	8
(Q)	Minimum value of quadratic expression corresponding to quadratic eqn obtained in above question occurs at x =	(2)	2
(R)	The sum of fifth power of roots of equation $f(x)=-x^3+x^2+1$ is given by K then $\left\|\frac{K}{5}\right\|$ is equal to	(3)	4
(S)	The no. of solutions of $f(\|x\|)=0$ is	(4)	3

Codes:

	P	Q	R	S
(A)	1	2	4	3
(B)	4	2	3	1
(C)	2	3	4	1
(D)	3	2	4	3

Space for Rough work

45. Let $b_1 = 50$ & $b_1, b_2, ...b_n$ be a sequence of numbers satisfying the rule $n(b_n - b_{n+1}) = n^3 + n^2 - b_n$. Then, match List (I) to List (II):

List –I		**List –II**	
(P)	$\frac{b_{16}}{16}$	(1)	-16
(Q)	b_{11}	(2)	50
(R)	b_{10}	(3)	-55
(S)	$\frac{b_{12}}{12}$	(4)	-70

Codes:

	P	Q	R	S
(A)	4	2	3	1
(B)	1	4	2	4
(C)	4	3	2	1
(D)	1	3	2	4

46. Let P be the plane parallel to y-axis & containing the points (1, 0, 1) & (3, 2, -1). Also $A \equiv (4,0,0)$ & $B \equiv (6, 0, -2)$ are 2 pts & $P(x_0, y_0, z_0)$ is a variable point on the plane $P = 0$

List –I		**List –II**	
(P)	If the equation of plane P = 0 is $x + ay + by = c$ then $\lvert a + b + c \rvert$ is	(1)	16
(Q)	If $PA + PB$ is minimum, then $\lvert 4x_0 + y_0 + 2z_0 \rvert$ is	(2)	12
(R)	If $\lvert PA - PB \rvert \in [0, \sqrt{N})$, then N is	(3)	3
(S)	If the reflection of the line AB in the plane P = 0 is $\frac{x-2}{1} = \frac{y-\alpha}{0} = \frac{Z+\beta}{-1}$, then $\alpha^4 + \beta^4$ is	(4)	8

Codes:

	P	Q	R	S
(A)	3	2	4	1
(B)	2	4	3	1
(C)	3	1	4	2
(D)	2	3	1	4

Space for Rough work

SECTION – D

(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

47. A game is played with fair cubic die which has one red side, two blue sides & three green sides. The result is the color of the top side after the dice is rolled. If the die is rolled repeatedly. The probability that second blue result occurs on or before the tenth roll can be expressed in the form of $\frac{3^p - 2^q}{3^r}$ where p, q, r are positive integers then the value of $p + r - q$ is _____

48. If area of triangle formed by the points $(2\alpha, \beta), (\alpha + \beta, 2\beta + \alpha)$ & $(2\beta, 2\alpha)$ is 8 sq. unit then area of triangle whose vertex are $(\alpha + \beta, \alpha - \beta), (3\beta - \alpha, \beta + 3\alpha)$ & $(3\alpha - \beta, 3\beta - \alpha)$ is _____

49. Let z_1, z_2, z_3 be complex numbers such that $|z_1| = |z_2| = |z_3| = 10$ & $z_2 \neq z_3$ then $\frac{\min\limits_{a \in R} |az_2 + (1-a)z_3 - z_1|}{|z_1 - z_2|\,|z_1 - z_3|}$ is equal to

50. Given circles C_1 & C_2 intersect at X & Y. Let l_1 be a line through the centre of C_1 & intersecting C_2 at points P & Q and let l_2 be the line through centre of C_2 intersecting C_1 at pts R & S. Let $C_1 : x^2 + y^2 - 2x - 2y + 1 = 0$ & $C_2 : x^2 + y^2 - 4x - 4y + 4 = 0$. If P, Q, R, S are concyclic with centre of circle passing through them is (α, β), then $\tan^{-1}(\alpha + \beta) + \tan^{-1}(\alpha + \beta - 1)$ is equal to $\tan^{-1}\left(\frac{10}{K}\right)$, then K is

51. Functions $P(x), Q(x), R(x)$ are differentiate on some open interval around O & satisfy the equations & initial conditions

$$P'(x) = 2P^2(x).Q(x).R(x) + \frac{1}{Q(x).R(x)}, P(0) = 1$$

$$Q'(x) = P(x)\,Q^2(x)\,R(x) + \frac{4}{P(x)\,R(x)}, Q(0) = 1$$

$$R'(x) = 3P(x)\,Q(x)\,R^2(x) + \frac{1}{P(x).Q(x)}, R(0) = 1$$

Then $P(x)\,Q(x)\,R(x) = \tan\left(nx + \frac{\pi}{4}\right)$, then n is equal to ___

Space for Rough work

52. If area of the region in the complex plane that consists of all points z such that $\frac{z}{40}$ and $\frac{40}{z}$ have real & imaginary parts lifting in the interval [0, 1] is A then $\frac{A}{100}$ is (take π = 3.14)

53. Suppose f is a function on the interval [1, 3] such that $-1 \le f(x) \le 1 \; \forall x$ & $\int_1^3 f(x)dx = 0$. If maximum value of $\int_1^x \frac{f(x)}{x} dx$ is $C_n\left(\frac{a}{b}\right)$ (where a, b are coprime numbers, then $a+b$ is equal to _____.

54. Let $y = g(x)$ is solution of differential equation $g'(x) + g(x) = \frac{2xe^{-x}}{1+g(x)e^{x}}$ such that $g(0) = 1$, then $\left[\frac{f(-1)}{e}\right]$ is equal to [where [.] denotes GIF]

Space for Rough work